Daily Geometry Drill #5

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Angle at the centre · Cyclic quadrilaterals · Tangent ⊥ radius · Power of a point · Touching circles: join the centres.

Section A Rapid Recognition

1. In a circle with centre O, the angle at the circumference subtending arc AB is $20°$. Write down the angle at the centre.

Ans: $40°$

2. AB is a diameter of a circle and P is any other point on the circle. Write down $\angle APB$.

Ans: $90°$

3. $ABCD$ is a cyclic quadrilateral with $\angle A = 70°$ and $\angle B = 85°$. Write down $\angle C$.

Ans: $110°$

4. The tangent at T to a circle makes an angle of $40°$ with the chord TA. By the alternate segment theorem, write down the angle in the alternate segment subtended by TA.

Ans: $40°$

5. Chords AB and CD of a circle intersect inside at P, with $PA = 2$, $PB = 6$ and $PC = 3$. Write down PD.

Ans: 4

6. A tangent from P touches the circle at T, and a secant through P cuts the circle at A then B. Given $PT = 3$ and $PA = 1$, write down PB.

Ans: 9

7. Write down the length of the tangent from $(7, 0)$ to $x^2 + y^2 = 36$.

Ans: $\sqrt{13}$

8. Two tangents from the point P to a circle of unknown radius make an angle of $60°$ and $OP = 6$. Write down the length of each tangent.

Ans: $3\sqrt{3}$

9. The two tangents from P to the circle $x^2 + y^2 = 25$ make an angle of $120°$. Write down OP.

Ans: $\frac{10\sqrt{3}}{3}$

10. In the cyclic quadrilateral $ABCD$, $\angle BAC = 30°$. Write down $\angle BDC$.

Ans: $30°$

Section B Manipulation Drills

1. A tangent at T makes an angle of 55° with chord TA. The angle in the alternate segment subtended by TA is?

A) 35°

B) 55°

C) 110°

D) 70°

Ans: B) 55°

2. $ABCD$ is cyclic with $\angle A = 2x$ and $\angle C = 3x$. Find x.

A) 24

B) 30

C) 36

D) 72

Ans: C) 36

3. A *tangential* (circumscribed) quadrilateral $ABCD$ has $AB = 5$, $BC = 7$ and $CD = 6$. By Pitot's theorem ($AB + CD = BC + DA$), find DA.

A) 4

B) 6

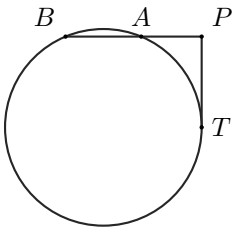
C) 8

D) 18

Ans: A) 4

4. From external P, tangent $PT = 12$ touches a circle and secant PAB meets it with $PA = 8$. Find the chord AB.

A) 6
B) 10
C) 12
D) 18



Ans: B) 10

5. From an external point P, two secants meet the same circle at P–A–B and P–C–D, with $PA = 4$, $PB = 18$ and $PC = 6$. Write down the length of the chord CD.

Ans: 9

6. The point P has $OP = 6$ with respect to a circle of radius 3. The angle between the two tangents from P is?

A) $30°$

B) $45°$

C) $60°$

D) $90°$

Ans: C) $60°$

7. Two circles of radius 6 each pass through the other's centre. They meet at A and B. How long is AB? *(after Primer 1.4 P16)*

A) 6

B) 12

C) $6\sqrt{2}$

D) $6\sqrt{3}$

E) $3\sqrt{3}$

Ans: D) $6\sqrt{3}$

8. Two concentric circles have radii in the ratio $2:5$. AC is a diameter of the larger circle, and the chord BC of the larger circle touches the smaller circle. If $AB = 8$, what is the radius of the larger circle? *(after Primer 1.4 P13)*

A) 16

B) 20

C) 8

D) 12

E) 10

Ans: E) 10

9. The line ST is a chord of the larger of two concentric circles and touches the smaller one. If $ST = 30$, find the area of the region between the circles. *(after Primer 1.4 P8)*

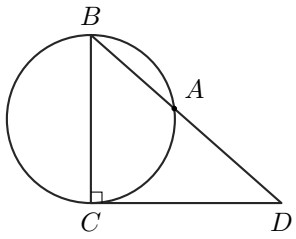
Ans: 225π

10. AB is a diameter of a circle and C is a point on the circle with $\angle ABC = 55°$. Write down $\angle BAC$.

Ans: $35°$

Section C Substitution & Structure

1. BC is a diameter of a circle and CD is the tangent at C. The line BD meets the circle again at A. If $AB = 7$ and $AD = 9$, what is the radius of the circle? *(after Primer 1.4 P40)*



A) 12
B) $4\sqrt{7}$
C) $2\sqrt{7}$
D) 8
E) 6

Ans: C) $2\sqrt{7}$

2. A regular octagon is inscribed in a circle with centre O and radius 2. A point X inside the circle is chosen so that the sum of the squares of its distances to the eight vertices is 40. What is OX? *(after SMC 2025 Q21)*

A) 1

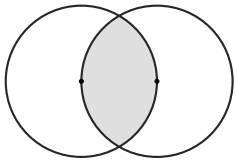
B) $\sqrt{5}$

C) 2

D) 3

E) $\sqrt{3}$

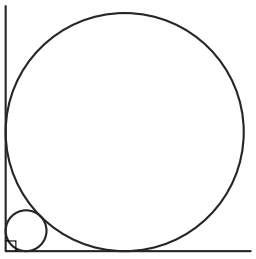
3. Two circles of radius 2 each pass through the other's centre. What is the area of the shaded region common to both circles? *(after Primer 1.4 P24)*



A) $\frac{4\pi}{3} - \sqrt{3}$
B) $\frac{8\pi}{3} - \sqrt{3}$
C) $\frac{4\pi}{3} + 2\sqrt{3}$
D) $\frac{8\pi}{3} - 2\sqrt{3}$
E) $2\pi - 2\sqrt{3}$

Ans: D) $\frac{8\pi}{3} - 2\sqrt{3}$

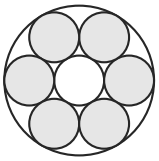
4. A circle of radius 1 touches both arms of a right angle. A larger circle touches both arms and the first circle. What is the radius of the larger circle? *(after Primer 1.4 P35)*



A) $1 + \sqrt{2}$
B) $2 + \sqrt{2}$
C) $2\sqrt{2}$
D) 3
E) $3 + 2\sqrt{2}$

Ans: E) $3 + 2\sqrt{2}$

5. Six equal discs lie between two concentric circles. Each disc touches its two neighbours and both circles. What fraction of the ring between the two circles is covered by the discs? *(after SMC 2016 Q21)*



A) $\frac{3}{4}$

B) $\frac{2}{3}$

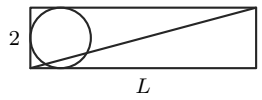
C) $\frac{7}{9}$

D) $\frac{1}{2}$

E) $\frac{3}{5}$

Ans: A) $\frac{3}{4}$

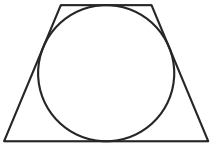
6. A circle of radius 1 touches three sides of a rectangle of width 2 and length L. A diagonal of the rectangle cuts the circle into two arcs, and the smaller arc is exactly a quarter of the circle. What is L? *(after SMC 2017 Q19)*



A) $4 - 2\sqrt{3}$
B) $4 + 2\sqrt{3}$
C) $2 + 2\sqrt{3}$
D) 2
E) $2 + 2\sqrt{2}$

Ans: B) $4 + 2\sqrt{3}$

7. All four sides of an isosceles trapezium touch a circle of radius 6, and the perimeter of the trapezium is 52. What is the difference between the lengths of its parallel sides? *(after SMC 2012 Q20)*

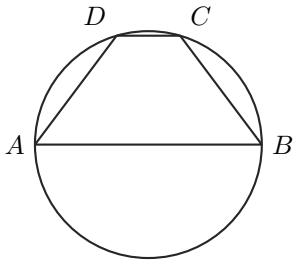


A) 12
B) 13
C) 26
D) 10
E) 5

Ans: D) 10

8. $ABCD$ is a cyclic quadrilateral in which AB is a diameter of the circle, $AB = 25$, $CD = 7$ and $AD = BC$. What is the length AD? *(after Primer 1.4 P33)*

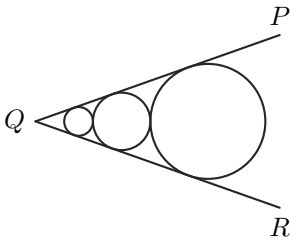
A) 12
B) 20
C) 16
D) $5\sqrt{7}$
E) 15



Ans: E) 15

Section D Challenge

1. Three circles each touch both arms QP and QR of an angle, and each touches the next. The largest radius is 4 times the smallest. What is $\cos\angle PQR$? *(after SMC 2011 Q24)*



A) $\frac{7}{9}$

B) $\frac{1}{3}$

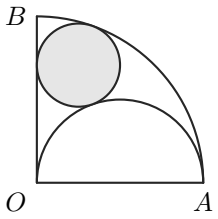
C) $\frac{2}{3}$

D) $\frac{8}{9}$

E) $\frac{4\sqrt{2}}{9}$

Ans: A) $\frac{7}{9}$

2. OAB is a quarter circle with radius 4. A semicircle is drawn on OA as diameter, inside the quarter circle. A circle touches OB, the semicircle and the arc AB. What is its radius? *(after SMC 2014 Q19)*



A) $\frac{4}{3}$
B) 1
C) $\frac{2}{3}$
D) $\sqrt{2}$
E) $\frac{3}{4}$

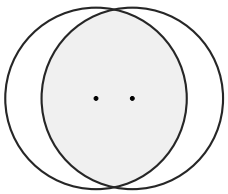
Ans: B) 1

3. A triangle is inscribed in a circle of radius 2. One of its angles is $45°$ and its area is $1 + \sqrt{3}$. What is its largest angle? *(after SMC 2021 Q22)*

A) $90°$
B) $120°$
C) $105°$
D) $135°$
E) $75°$

Ans: C) $105°$

4. Two circles of radius 5 have centres 2 apart. Let S be the largest square that fits in the region common to both circles with its sides parallel to the line of centres, and T the largest that fits with its diagonals along and perpendicular to that line. What is the ratio of the area of S to the area of T? *(after SMC 2019 Q25)*



A) 1

B) $\dfrac{9}{16}$

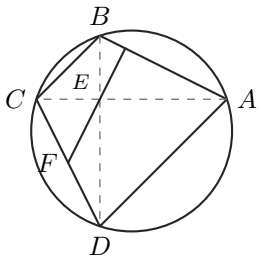
C) $\dfrac{8}{9}$

D) $\dfrac{9}{8}$

E) $\dfrac{3}{4}$

Ans: D) $\frac{9}{8}$

5. $ABCD$ is a cyclic quadrilateral whose diagonals are perpendicular and meet at E. The line through E perpendicular to AB meets CD at F. Prove that F is the midpoint of CD.



Ans: Proof: see method